



Final Year Project Long Proposal

Cloud Policy Cost and Security Simulator

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Revision History

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1. Project Description

The proposed system will be a "Cloud Policy, Cost and Security Simulator," a revolutionary "Digital Twin" platform designed to master the complexities of cloud governance, financial operations (FinOps), and cybersecurity through advanced intelligence. Unlike standard management tools that connect to live, expensive production environments, this system creates a completely isolated, high fidelity virtual ecosystem. It integrates cutting edge modules such as autonomous policy enforcement, predictive cost forecasting, and AI driven threat defense within a safe, risk free sandbox.

The Cloud Policy, Cost and Security Simulator operates as a sophisticated training ground that transforms theoretical knowledge into operational mastery. It moves beyond basic monitoring tutorials to offer a predictive and autonomous environment where users can witness the intersection of DevOps, Security, and AI without the fear of "bill shock" or data breaches. The platform is a high stakes simulator that provides students and SMEs with the opportunity to experiment with complex cloud strategies such as mitigating DDoS attacks, optimizing enterprise budgets, or enforcing governance compliance using simulated resources and synthetic data.

The system creates a "Digital Twin" of a cloud infrastructure. Users interact with this virtual twin to manage simulated resources (Virtual Machines, Databases) that behave exactly like real cloud instances. They can visualize budget impacts via machine learning models trained on synthetic historical data, and detect zero day threats in real time using AI agents monitoring synthetic network traffic. This creates a sophisticated, secure, and hyper efficient learning environment for the next generation of cloud engineers and AIOps specialists, specifically tailored to address the skills gap in the local market.

The core innovation lies in the Synthetic Data Generator. By generating massive datasets of realistic cloud usage, cost anomalies, and security threats, the system powers an Intelligent Governance Engine that translates human policies into executable code, a Predictive FinOps module that forecasts costs using Time-Series algorithms, and an AI-Driven Security suite that identifies anomalies using Deep Neural Networks. The platform allows users to provide virtual resources, automatically enforce compliance, visualize risk scores using dynamic heatmaps, and oversee autonomous remediation agents that fix vulnerabilities instantly.

2. Real World Problem

The increasing migration of organizations to cloud environments has introduced greater infrastructure complexity, rising costs, and elevated security risks. However, the expertise required to manage these complex autonomous cloud operations (AIOps) is currently lacking within Pakistan's workforce. There is a significant skills gap between Subject Matter

Experts (SMEs) and students, who often do not possess sufficient practical experience to operate in complex cloud environments without suffering data loss, financial inefficiency, or security breaches.

The root of this problem is educational. Conventional modes of learning are largely passive and lecture based, failing to respond to the dynamic nature of modern cyber threats and financial governance in the cloud. Furthermore, educational frameworks remain reactive and manual while the industry shifts rapidly towards Autonomous Cloud Operations. The consequences for organizations are dire: security misconfigurations, uncontrolled billing, and regulatory non compliance.

Most critically, the lack of a safe, simulation-based training environment impacts both the education sector and local industry. Graduates enter the workforce with theoretical concepts but zero exposure to managing AI driven Security Operations Centers (SOCs) or FinOps strategies. SMEs hesitate to adopt cloud technologies due to the valid fear of "bill shock" (unexpected high costs) and cyber attacks, knowing their staff lack the expertise to predict or mitigate these risks. Without a safe environment to practice "Self-Healing" operations and "Predictive Governance" without incurring real costs, the local industry cannot scale efficiently, and the skills gap continues to widen, preventing Pakistan from competing at global standards.

The Cloud Policy, Cost and Security Simulator addresses these critical issues by deploying a comprehensive, AI powered simulation platform. It bridges the gap between theory and practice by simulating a full-scale cloud environment where all consequences are virtual. The Governance Engine enforces best practices automatically, the Predictive FinOps module teaches financial foresight using synthetic time series data, and the AI Security module recreates actual cyber threats in a safe environment to demonstrate how Autonomous Agents defend the infrastructure. This transforms passive learning into a high impact, active experience, enabling users to master cloud complexities with confidence.

3. Project Stream

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|--|--|
| ☒ Web-Based Platform | ☐ Desktop Application |
| ☐ Mobile App | ☐ Game |
| ☐ Hardware-Based | |

4. Modules

The project has been designed to be divided into phases that are high impact and have their own designated phases to make sure that every member of this team has an equally balanced workload.

Table 1: Module Breakdown and Assignment

Deadline	Modules	Student
P1 Mid	Data Generator	Maryam Shahzad (2212396)
	Organization Manager	M. Abdur Rehman Khan (2212399)
	Authentication	Wasfa Nauman Bhatti (2312462)
P1 Final	Resource Viewer	Wasfa Nauman Bhatti (2312462)
	Resource Simulator	M. Abdur Rehman Khan (2212399)
	Threat Detector	Maryam Shahzad (2212396)
P2 Mid	Report Generator	Wasfa Nauman Bhatti (2312462)
	Policy Engine	M. Abdur Rehman Khan (2212399)
	Remediation Agent	Maryam Shahzad (2212396)
P2 Final	User Settings	Wasfa Nauman Bhatti (2312462)
	Cost Forecaster	M. Abdur Rehman Khan (2212399)
	Audit Trail	Maryam Shahzad (2212396)

5. Names of existing similar FYPs (if repeated)

Cloud Management System 2020

Cloud Cost Estimator 2019

6. New Functionalities (main) to be added Features

The Cloud Policy, Cost and Security Simulator introduces sophisticated, industry grade features tailored for a simulation environment. These features revolutionize cloud training by introducing intelligence, foresight, and automation without the risks associated with live production environments.

Simulated Resource Utilization Monitoring: To address the critical need for understanding resource performance, the platform generates high fidelity synthetic utilization metrics. For every simulated resource (VM, Database), the system creates realistic data points for CPU load percentage, Memory consumption (GB%), Disk I/O, and Network throughput. This allows users to practice "Rightsizing" (optimizing resources) and identifying "Wastage" (resources running with low utilization) within a risk free sandbox. This functionality bridges the gap between theoretical cloud concepts and practical infrastructure management.

Intelligent Governance Engine (NLP): The system empowers users to write governance policies in plain English (e.g., "All databases must be encrypted and tagged with 'Finance'"). The NLP engine parses this text and automatically compiles it into executable simulation rules. This teaches the vital link between human-readable policy and machine enforced code.

Predictive FinOps & Resource Optimization: Instead of just showing current bills, the system looks into the future. Using Time Series Forecasting algorithms on the synthetic historical data, it analyzes the user's simulated resource provisioning history to predict future costs with high accuracy. It utilizes clustering algorithms to detect "wasteful" idle resources based on their utilization patterns and suggests automated optimizations to reduce spending.

AI Driven Security & Anomaly Detection: This is the core "Wow" factor. A Deep Neural Network (DNN) monitors simulated network traffic logs in real time, distinguishing between normal operational behavior and sophisticated cyber threats like DDoS or brute force attacks. It provides a live view of the security posture and utilizes the synthetic utilization data to detect threats (e.g., identifying a DDoS attack by a sudden spike in simulated CPU and Network I/O).

Autonomous Remediation Agent: The Cloud Policy, Cost and Security Simulator features a Reinforcement Learning inspired agent. This agent monitors the cloud state and, upon detection of a critical security misconfiguration (e.g., a public database) or resource wastage (e.g., a VM running at 0% CPU for days), the AI computes a risk score and automatically implements a remediation plan. This models a "Self-Healing" cloud environment where the infrastructure patches itself automatically without human intervention.

7. Development Environment

Tools: VS Code, GitHub, Figma.

Frontend: React.js, Recharts, Tailwind.

Backend: Python (Flask/Django).

Database: PostgreSQL + Redis.

AI/ML: Libraries (Scikit-Learn, TensorFlow/Keras, Pandas, NumPy).

8. Introduction

The global landscape of cloud computing has evolved rapidly from simple virtualization to complex, AI driven ecosystems. However, the educational infrastructure required to train professionals for this new era is severely lacking, particularly in developing regions like Pakistan. Students and SMEs are forced to choose between expensive, high risk production environments (where a mistake costs real money) or overly simplified tutorials

that offer no real insight into resource utilization or autonomous defense.

There is a critical disconnect between the theory taught in universities and the practical, autonomous defense mechanisms demanded by the industry (AIOps). This proposal addresses this challenge by engineering a high fidelity "Digital Twin" of the cloud. It is not merely a monitoring tool; it is a living simulation where users can interact with intelligent agents.

The platform integrates advanced Machine Learning to demonstrate how governance, cost management, and security are interconnected. By offering a sandbox where users can launch simulated attacks to see how the AI responds, or over provision resources to see how costs spike based on utilization metrics, the project provides a visceral, risk free learning experience. It serves as the foundation for cultivating the next generation of Cloud Architects, FinOps specialists, and Security Engineers, specifically tailored to close the local skills gap.

9. Application Review

We analyzed industry standard cloud management tools and educational platforms to identify critical gaps that this project fills. The focus was on the level of automation, the depth of security features, and the learning curve associated with each tool. The analysis reveals a clear opportunity for a localized, intelligent platform that combines governance, cost, and security into one cohesive simulation.

AWS Cost Explorer & Azure Advisor: It operates as the industry standard for managing enterprise clouds. These platforms offer deep insights into spending patterns, security configurations, and resource utilization. They provide features like budget alerts, reserved instance recommendations, and security scorecards. Despite their powerful capabilities, these tools require active, paid cloud subscriptions. They are geared towards production set ups and this means users will incur costs whenever they make a mistake. It does not have a sandbox mode, where the students can test DDoS attacks or misconfigurations without a real life impact. Moreover, the interface and complexity of the data are too complex to the novice. [1].

The key characteristics of the application are presented below:

- **Real-Time Cost Tracking:** provides users with detailed breakdowns of service, region, and type of usage spending.
- **Security Recommendations:** Automated scanners provide solutions to security vulnerabilities, including unencrypted storage or open port.
- **Budget Management:** It enables the establishment of individual budgets, and also gives alerts when the spending nears a specified limit.

- **Resource Optimization:** Algorithms will propose rightsizing resources (e.g. use cheaper instance types) to use less money.
- **Usage Forecasts:** Basic prediction tools estimate future costs based on current month-to-date spending

The limitations of the application are listed below:

- **Financial Barrier:** Usage of these tools is tied to active billing accounts. Experimentation results in actual financial charges.
- **High Complexity:** The terminology and vast amount of data presented assume a high level of prior cloud expertise.
- **No Simulation Environment:** Users cannot "reset" the environment to undo changes. Mistakes must be manually fixed, which can be costly.
- **Lack of Educational Context:** The tools show what is wrong but often fail to explain why or teach the user how to prevent it.
- **Limited AI Integration:** While they offer recommendations, they lack autonomous agents that can automatically execute fixes

Azure Sandbox / AWS Educate: It operates as a limited-time access program providing credits for students. It aims to lower the barrier to entry by offering free credits for educational purposes. While it provides access to real consoles, the credits are limited and expire quickly. The focus is on provisioning resources, not on the governance or security management of them. There is no integration of AI to demonstrate autonomous operations or advanced threat hunting within the provided labs. [2].

The main features of the Application are listed below:

- **Free Credits:** Provide a starting balance for students to use real cloud services without immediate payment.
- **Standard Console Access:** Users gain access to the actual AWS or Azure interfaces used by professionals.
- **Pre-built Labs:** Guided tutorials walk users through basic tasks like launching a website or storing a file.

The limitations of the application are listed below:

- **Credit Depletion:** Once credits are used up, the account is frozen, preventing further learning.

- **Possibility of a Bill Shock:** New users can easily end up with their resources being used on costly services (e.g. GPU instances) that run in the background and empty their wallets immediately.
- **No Governance:** For the students, the system does not prevent them to create resources using random names, no tags, and insecure settings.
- **Static Content:** Static content labs can be either a video or a step-by-step guide that is not responsive.

Table 2 provides a side-by-side evaluation of the platforms listed above.

Table 2: Applications Comparison

Features	Applications		
	AWS/Azure Tools [1]	Azure/AWS Educate [2]	Proposed Simulator
Real Time Cost Tracking	✓	✗	✓
Governance Enforcement	✓	✗	✓
Risk Free Simulation	✗	✓	✓
AI Driven Threat Detection	✗	✗	✓
Autonomous Remediation	✗	✗	✓
Resource Utilization Metrics	✓	✗	✓
Deep Learning Education	✗	✗	✓
Educational Focus	✗	✓	✓
Financial Barrier	✗	✗	✓

10. Problem Statement

Current cloud management education fails to provide a holistic, intelligent learning environment that balances theory with practice. Students and professionals are left to navigate the complexities of Governance, FinOps, and Security in isolation, often using tools that are either too dangerous (production environments with real costs) or too simple (static tutorials).

There is a distinct lack of platforms that simulate the interconnected nature of cloud operations specifically, how a security misconfiguration leads to a compliance failure and

a cost spike based on resource utilization. The lack of Autonomous features implies that users are not being trained for the AIOps era. This creates a reactive workforce instead of a proactive one, leading to security weaknesses and financial inefficiency in the practical sense.

Furthermore, the concept of Resource Utilization is often taught theoretically. Students rarely get hands on experience analyzing CPU, Memory, and I/O bottlenecks because doing so on real cloud servers requires sustained, expensive provisioning. Additionally, the challenge of modeling storage utilization costs without actual hardware constraints is rarely addressed in curricula. Our challenges—specifically the need to consider resource utilization highlight the requirement for a simulated environment where these metrics (CPU, RAM, Disk I/O) can be generated, manipulated, and visualized without physical hardware constraints.

The challenges identified directly inform the design of this proposed solution. By integrating intelligent automation and synthetic data generation, the system solves the problem of manual error and passive monitoring. The inclusion of predictive security addresses the gap in threat preparedness. The unified governance module solves the issue of fragmented learning. The platform provides a cohesive environment where users can experience the full lifecycle of cloud operations—from provisioning and utilization monitoring to autonomous defense empowering them to become proactive, strategic thinkers in the cloud domain.

11. Scope

The Cloud Policy, Cost and Security Simulator aims to revolutionize cloud literacy and infrastructure management through an intelligent, AI-enabled "Digital Twin" Simulation. It features self governing enforcement of governance, predictive financial forecasting, AI based threat detection, and self healing security agents. The platform has two user levels: Learners/Admins and Organization Managers. A risk free simulated complex attack environment, governance rule testing, and real time cost impact visualization become available to the learners. Administrators control the access of multi tenants, monitor automated remediation policies, and oversee deep learning security logs.

Crucially, the scope is defined as a simulation:

In Scope: Creating, managing, and terminating virtual cloud resources (VMs, DBs); generating and visualizing synthetic resource utilization metrics (CPU, RAM); running AI threat detection on synthetic network traffic; forecasting costs based on simulated historical data.

Out of Scope: Access to real production cloud accounts (AWS/Azure); incurring real financial charges; physical hardware management; native mobile applications; blockchain

functionality.

The solution comprises the following core modules, each with its specified feature set:

Module: Authentication & User Interface

- Register/Login with Email Verification
- User Profile Management UI
- Password Recovery Interfaces

Module: Organization & Access Control

- Organization and Tenant Creation
- User Invitation System
- JWT Token Management
- Access Rights Configuration

Module: Resource Simulation Engine

- Provision Virtual Machines and Databases (Simulated)
- Dynamic IP Allocation (Virtual Networking)
- Simulated Resource Status & Utilization Monitoring
- Resource Status Management (Running, Stopped, Failed)
- Bulk Actions (Tagging, Deletion)

Module: Dashboard & Resource View

- Interactive Dashboard Layout
- Real time Utilization Heatmaps (CPU/Memory usage)
- Resource Inventory Tables
- Quick Status Overviews
- Sidebar and Navigation Menus

Module: Reporting & Visualization

- Historical Utilization Trend Charts
- Cost Trend Analysis Charts
- Interactive Risk Heatmaps
- PDF Report Generation

- Data Export Capabilities (CSV)

Module: Governance & Policy Engine

- Natural Language Policy Parser (NLP)
- Naming Convention Validator
- Automated Tagging Enforcement
- Utilization Based Policies
- Compliance Scoring Algorithm

Module: Settings & Preferences

- Dashboard Customization (Widgets)
- Notification Preferences (Email/In-App)
- Security Settings (2FA Configuration)
- Theme Selection (Dark/Light Mode)

Module: Synthetic Data Generator

- Simulated Network Traffic Scripting
- Synthetic Utilization Metric Generation
- Log Generation (Normal vs Attack)
- Data Balancing for Training
- Dataset Export for AI

Module: AI Driven Anomaly Detection

- Deep Neural Network (DNN) Model Training
- Real-Time Traffic Classification
- Anomaly Detection based on Utilization Spikes
- Threat Visualization Dashboard
- Attack Scenario Selection (DDoS, Brute Force)

Module: Autonomous Remediation Agent

- Auto-Remediation Rules Configuration
- Automated Firewall Blocking (Simulated)
- Auto Scaling/Shutdown based on Utilization

- Admin Override for Critical Actions

Module: Predictive FinOps Logic

- Real-Time Cost Calculation Engine
- Budget Guardrails and Threshold Logic
- Time-Series Cost Forecasting (AI Powered)
- Resource Wastage Detection Logic

Module: Audit & Compliance Trails

- Detailed Action Logs (Who did What)
- Security Incident History
- Utilization Audit Logs
- Compliance Report Templates
- Log Archival and Retrieval

12. Significance

The successful implementation of the Cloud Policy, Cost and Security Simulator will deliver substantial benefits to the educational and industrial sectors of Pakistan.

For students, the platform bridges the massive gap between academic theory and industry practice. It offers a safe space to master AIOps, Security, and Resource Utilization Optimization without the fear of causing financial damage or security breaches. This significantly boosts employability and technical confidence by providing experience with tools that behave like enterprise grade systems.

For SMEs, it provides a "Test Bed" to train staff and forecast costs without financial exposure. It allows small businesses to understand the nuances of cloud billing and governance before committing to a real cloud provider.

On a larger industrial scale, the project speeds up the process of adoption of digital technologies by demystifying cloud security, governance, and utilization management. The platform makes the local market able to compete internationally by creating a culture of intelligent and independent cloud management, minimizing the risks of cyber threats and operational inefficiency through the use of advanced AI driven simulations.

13. Tools and Techniques

The system has a powerful, state of the art technology stack that is streamlined into high-performance simulation and smooth integration of AI. The modules are driven by

specialized frameworks that are selected by their industry supremacy and ability. The engine of governance utilizes the principles of Natural Language Processing, whereas the security package utilizes Deep Learning libraries. The dashboard is based on powerful visualization libraries that dynamically display intricate data. Combined, these technologies can be used to form a stable, scalable, and intelligent platform, which resembles the behavior of enterprise-grade cloud systems.

- **Frontend Management:** React.js is an application built on the dynamic, single page application (SPA) of the responsive architecture. Tailwind CSS will bring a professional look, but indeed modern, whereas Recharts and Chart.js will make it possible to display the complex data visualizations of cost trends and risk heatmaps. Axios handles effective communication of API between the frontend and the backend.
- **Backend Tools:** The backend is developed in Python with Flask framework because it is lightweight and easily integrates with AI libraries. It handles server logic, session management and routing. JWT is used to implement authentication, Pandas and NumPy are employed to process the data and generate synthetic data backends.
- **Database Storage:** PostgreSQL is the main relational database where there are user profiles, simulated cloud resources, policy configurations, and security logs. It provides data integrity and helps in complex queries required to provide the simulation engine.
- **AI Techniques:** Scikit-Learn and TensorFlow/Keras are used in the system to construct Machine Learning models. In particular, it applies the Random Forests and Deep Neural Networks to anomaly detection, Linear Regression/LSTM to cost forecasting, and rule-based logic along with NLP concepts to governance engine. The models are trained on artificial training datasets that are generated by Synthetic Data Generation methods.

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